

OLMo SIH / ICL Developmental Side Experiment

Completed behavioral and RI developmental results, methodology comparison, interpretation, and follow-up checks — September 24, 2026

Side question: does OLMo-2-1124-7B develop few-shot ICL behavior during the same training phase in which SIH-like Relation Index structure appears? The broad behavioral boundary (step 600 to step 1000) was established before interpreting RI timing. After the RI sweep revealed a routing maximum near step 850, we ran a targeted behavioral follow-up at steps 700, 850, and 900 to localize the transition more precisely. The report therefore distinguishes the independently established broad window from this denser, hypothesis-motivated interpolation.

1. Frozen behavioral experiment

Dataset. Four synthetic classification tasks were evaluated at 0, 1, 2, 3, 4, 5, 10, and 20 shots. The frozen assessment contains 50 examples per task-shot condition, or 1,600 examples per checkpoint. Labels 0–8 are single OLMo tokens. The same prompt IDs are reused across training checkpoints, enabling paired checkpoint comparisons.

Checkpoints. Seventeen OLMo Stage-1 checkpoints were evaluated. The original 14-checkpoint schedule was augmented with steps 700, 850, and 900 to resolve the early transition. Because OLMo’s token labels are rounded, step 600 and step 700 are both labeled 3B, while steps 850 and 900 are both labeled 4B; all analysis therefore uses the exact revision/optimizer step as the checkpoint identity.

Tokens	Step	Tokens	Step	Tokens	Step
1B	150	5B	1,000	80B	19,000
3B	600	9B	2,000	160B	38,000
3B	700	13B	3,000	319B	76,000
4B	850	17B	4,000	638B	152,000
4B	900	26B	6,000	1498B	357,000
		38B	9,000	3896B	928,646

Metrics. Constrained accuracy and gold-vs-best-wrong margin measure task performance; vocabulary-wide format validity and legal-label probability mass distinguish response-format acquisition from actual pattern discovery. The analysis also uses 20-shot minus 1-shot accuracy gain, Wilson confidence intervals, and exact paired McNemar/binomial tests across checkpoints.

2. How our ICL and RI measurements correspond to Ren et al.

The side experiment is designed as a **close operational adaptation**, not an exact reproduction, of Ren et al. The central definitions—few-shot format compliance, few-shot pattern discovery, and Relation Index—are preserved as far as possible, while the implementation is adapted to OLMo-2 and augmented with controls that separate effects which are combined in the original measurements.

ICL: what is the same

Ren et al. evaluate few-shot ICL with four synthetic classification tasks: two binary tasks using fruit/month and furniture/profession entity sets, a four-class fruit/month task, and a nine-class fruit/animal/month task. Their prompts contain demonstrations followed by a query, and when the number of demonstrations is large enough they ensure that every class is represented. They distinguish **format compliance**—producing a one-token numerical answer in the expected format—from **pattern discovery**—predicting the correct label implied by the demonstrations.

Our behavioral stream preserves these core choices. We use the same four task families, the same shot ladder {0, 1, 2, 3, 4, 5, 10, 20}, and enforce demonstration-class coverage whenever the number of shots permits it. Every label 0–8 is verified to be a single OLMo token. The assessment uses 50 frozen examples per task-shot condition, and the exact same prompt IDs are reused at every OLMo checkpoint. This makes checkpoint-to-checkpoint comparisons paired rather than relying on newly sampled prompts.

ICL: measurement differences

The main difference is how we separate *format* from *mapping accuracy*. Ren et al. force a one-token generation; their format score asks whether the generated token has the expected numerical form, and their pattern-discovery accuracy asks whether that generated token is the correct label. Consequently, a format failure also becomes a pattern-discovery error.

Our evaluator instead records both constrained and vocabulary-wide next-token behavior. For the primary pattern-discovery score we compare only the legal task-label logits and choose

$$\hat{y}_{\text{constrained}} = \arg \max_{y \in \mathcal{Y}_{\text{task}}} z_y,$$

then report constrained accuracy and the gold-minus-best-wrong logit margin. Separately, we inspect the full vocabulary: whether the global top token is a legal label, whether it is the gold label, the total probability mass on legal labels, and the gold-label probability. Thus our headline task accuracy is intentionally **less confounded by output-format failure** than the paper’s generated-label accuracy, while our separate format metrics preserve the format-compliance question.

Our format-valid criterion is also somewhat stricter: it requires the vocabulary-wide top token to be one of the task’s *legal label tokens*, not merely a numerical-looking token. Because of these differences, absolute accuracy values should not be treated as numerically interchangeable with Ren et al.; the strongest comparison is the **developmental ordering and timing** of format acquisition versus pattern discovery.

We also do not reproduce two other parts of Ren et al.’s ICL suite in this side experiment: their loss-reduction measurement and their relation-justification tasks. Our behavioral analysis is restricted to the four classification tasks that most directly define the paper’s few-shot pattern-discovery curves.

ICL component	Ren et al.	Our OLMo adaptation
Task families	Two binary, four-class, nine-class synthetic classification tasks	Same four classification task families
Shots	Few-shot curves including 0–5, 10 and 20 shots	Same $\{0, 1, 2, 3, 4, 5, 10, 20\}$ grid
Class coverage	Each class represented when shots exceed class count	Same rule, audited before evaluation
Format compliance	One generated token; numerical format counted as correct format	Global next-token argmax must be a legal label; also record legal-label probability mass
Pattern discovery	Correct generated label	Primary metric is constrained accuracy among legal labels; unconstrained correctness is also stored
Checkpoint comparison	Developmental checkpoints of the paper’s InternLM training run	Frozen prompts evaluated across OLMo Stage-1 checkpoints, enabling paired tests
Additional metrics	Accuracy curves	Margin, Wilson intervals, shot gains, paired transition tests, label mass and gold probability

RI: paper definition and our implementation

For a semantic triplet $T = (t_s, \text{relation}, t_o)$ and a current token position $j \geq \max(s, o)$, Ren et al. first identify heads whose QK circuit attends to the source token. In the standard RI analysis they require both

$$s = \arg \max_{k \leq j} A_{j,k}^h \quad \text{and} \quad \frac{A_{j,s}^h}{\max_{k \neq s} A_{j,k}^h} > \tau, \quad \tau = 2.2.$$

They then project the head’s raw-embedding OV output to the vocabulary,

$$z^{h,j} = x_j W_V^h W_O^h W_U, \quad p^{h,j} = \text{softmax}(z^{h,j}),$$

extract probabilities for the unique tokens already visible in the context, subtract their mean and clamp negative values,

$$q_{t_k}^{h,j} = \max\left(0, p_{t_k}^{h,j} - \mathbb{E}_{k' \leq j}[p_{t_{k'}}^{h,j}]\right),$$

and define the Relation Index contribution

$$a_T^{h,j} = \frac{q_{t_o}^{h,j}}{\sum_{k \leq j} q_{t_k}^{h,j}}.$$

The head’s RI is obtained by averaging these contributions over eligible current positions and triplets.

For developmental checkpoints, Ren et al. explicitly **drop the $\tau = 2.2$ dominance requirement** because early attention is too diffuse, retaining only the condition that the source is the attention argmax. Our primary developmental RI follows that same relaxed eligibility rule. We simultaneously compute the original $\tau = 2.2$ version as a sensitivity analysis rather than using it to define the main early-training curve.

RI: what is the same and what is adapted

We use AGENDA semantic relations and the same seven relation categories: Part-of, Compare, Used-for, Feature-of, Hyponym-of, Evaluate-for, and Conjunction. As in the paper, entities are replaced by single letters so that source and target are single-token anchors. The primary OV computation uses the **raw current-token embedding**, matching the paper’s RI construction, and only positions after both source and target are eligible.

Several adaptations are important:

- **OLMo-safe preprocessing.** We require unambiguous single occurrences and tokenizer-safe single-letter anchors under the OLMo tokenizer. Sixty degenerate rows in which source and target collapse to the same entity/token are preserved in an exclusion manifest but omitted from RI. The final developmental assessment is deterministically balanced at 100 eligible triplets for each of the seven relations (700 triplets total).
- **Function-word filtering differs.** Ren et al. remove frequent function words identified by spaCy after entity replacement. Our frozen OLMo-safe stream does not apply that exact filtering step. We do not modify the frozen data retroactively; this is recorded as a methodological deviation and can be tested later as a sensitivity analysis.
- **Architecture-correct QK gate.** Rather than applying a simplified embedding-only QK expression, we use OLMo’s actual eager attention probabilities for the argmax/dominance test. This preserves OLMo-specific normalization and RoPE while retaining the paper’s semantic eligibility criterion: “does this head route from j to the source token?”
- **Exact context-only OV optimization.** The paper describes a full-vocabulary softmax and then extracts probabilities of visible context tokens. We compute logits only for those unique visible token IDs. This is mathematically exact for Equation (3), not an approximation: the common full-vocabulary softmax normalization factor cancels after mean centering and in the final ratio. A numerical equivalence test is included in the sweep notebook.
- **Forward direction is primary.** The paper also reports reverse semantic triplets. Our completed primary sweep uses the forward head→tail direction; reverse relations remain a possible sensitivity analysis.
- **Extra diagnostics.** In addition to conditional RI, we separately record how often the source passes the QK argmax gate, the strict- τ frequency, RI per source-routing opportunity, and two null-target controls. These quantities are not substitutions for the paper’s RI; they are additional measurements used to determine whether developmental change comes from *routing frequency* or from *conditional OV selectivity*.
- **Head stability.** We scan all 1,024 OLMo heads at every selected checkpoint. In addition to per-checkpoint top heads, we freeze high-RI coordinates at the final checkpoint and trace those same coordinates backward, with top- k Jaccard and all-head rank stability. This is an added guard against interpreting a different post-hoc head set at every checkpoint as one continuously developing mechanism.

RI component	Ren et al.	Our OLMo adaptation
Dataset	AGENDA semantic relations; entities replaced by letters	Same relation family and letter reduction; OLMo tokenizer-safety filtering and balanced 700-triplet assessment
Developmental QK gate	Source token is attention argmax; strict τ dropped for early checkpoints	Same relaxed argmax gate is primary; $\tau = 2.2$ retained as sensitivity
OV score	Raw embedding $x_j W_V^h W_O^h W_U$, vocabulary softmax, mean-center visible-token probabilities	Same raw-embedding OV quantity; algebraically exact context-only implementation
Text preprocessing	Frequent spaCy function words removed	Exact spaCy removal not used; documented deviation
Direction	Forward and reverse semantic relations reported	Forward direction in the primary completed sweep
Aggregation	Average RI over eligible positions/triplets	Conditional RI plus separately stored QK frequency and RI-per-opportunity
Head analysis	Developmental RI curves and highlighted heads	All 1,024 heads plus fixed-final-head backward traces and stability diagnostics

Interpretation: The experiment is therefore comparable to Ren et al. at the level most important for the research question—the same toy ICL task families, the same format-versus-pattern distinction, and the same developmental RI eligibility and OV logic—but it is not an exact replication. The largest differences are our constrained primary ICL accuracy, OLMo-specific/token-safe AGENDA preprocessing, omission of the paper’s spaCy function-word filter and reverse direction in the primary RI sweep, and architecture-correct QK evaluation. These differences should be stated whenever absolute values are compared; developmental timing and qualitative ordering are the safer cross-model comparisons.

3. Main behavioral result: the transition localizes to steps 700–900 (approximately 4B tokens)

The three added checkpoints substantially narrow the behavioral result. At step 600 and step 700, mean 20-shot accuracy is 0.360 and 0.335 respectively, the mean 20-shot minus 1-shot gain is non-positive, and none of the four tasks has a 95% Wilson lower confidence bound above its task-specific chance level. At step 850, mean accuracy rises to 0.460, the mean high-shot gain becomes positive (+0.060), and two of four tasks have confidence intervals entirely above chance. At step 900, mean accuracy reaches 0.555, all four tasks have Wilson lower bounds above chance, and every task has a positive 20-shot minus 1-shot gain. By step 1000, mean accuracy rises further to 0.635 and the mean high-shot gain reaches +0.240.

Thus the best current localization is:

transition begins between steps 700 and 850 and is robust across tasks by step 900.

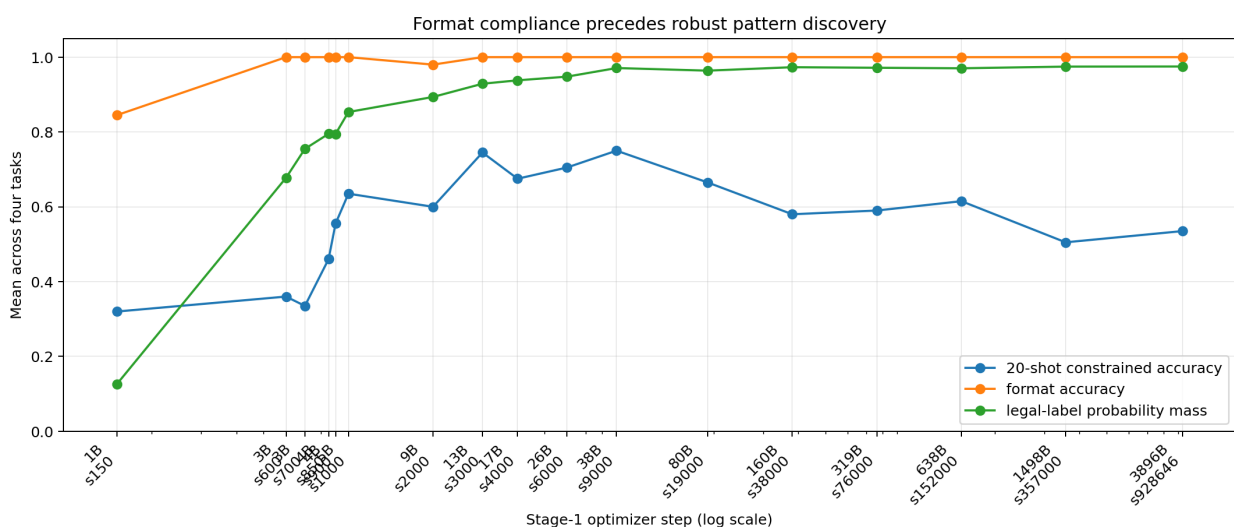
In rounded token terms, the main behavioral transition is centered near **4B training tokens**. The stricter operational landmark “all four 20-shot Wilson intervals above chance” is first satisfied at **step 900**.

Task	Chance	s600	s700	s850	s900	s1000	First CI > chance
Fruit/month	0.50	0.54	0.50	0.64	0.68	0.90	4B / s850
Furniture/profession	0.50	0.48	0.48	0.60	0.72	0.84	4B / s900
Four-class	0.25	0.26	0.24	0.36	0.54	0.52	4B / s900
Nine-class	0.111	0.16	0.12	0.24	0.28	0.28	4B / s850

Step	Mean 20-shot	Mean 20–1 gain	Tasks CI > chance
600	0.360	-0.010	0 / 4
700	0.335	-0.020	0 / 4
850	0.460	+0.060	2 / 4
900	0.555	+0.155	4 / 4
1000	0.635	+0.240	4 / 4

The dense 700/850/900 checkpoints were added as a follow-up after the RI routing peak had been observed. Therefore, the original 600–1000 behavioral window is independent of the RI result, while the exact 850/900 localization should be treated as targeted follow-up evidence rather than a preregistered temporal prediction.

4. Format learning still precedes pattern discovery



Mean 20-shot format accuracy is already 1.00 at steps 600 and 700 and remains 1.00 through steps 850, 900, and 1000. Yet mean task accuracy is only 0.360 at step 600 and 0.335 at step 700. Legal-label probability mass also rises before robust mapping accuracy: approximately 0.678 at step 600, 0.755 at step 700, 0.795 at steps 850/900, and 0.853 at step 1000. This preserves the original qualitative conclusion: OLMo learns the response format before it reliably exploits the demonstrations to recover the arbitrary label mapping.

Interpretation: The denser sweep strengthens, rather than weakens, the format-versus-pattern distinction. Output-format behavior is already essentially solved before the sharp improvement in few-shot mapping performance around steps 850–900.

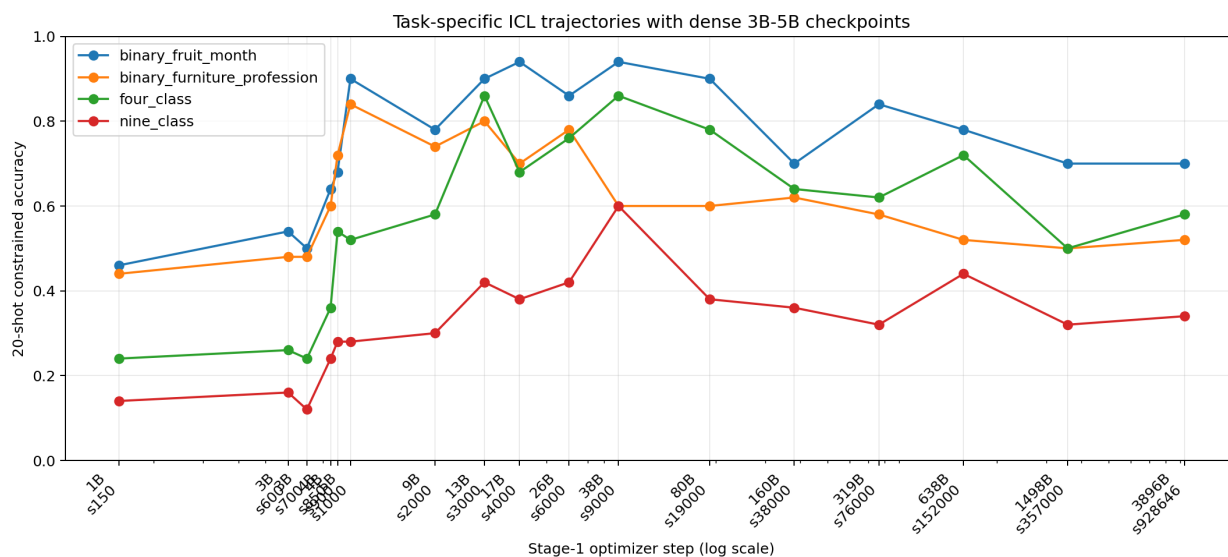
5. Paired evidence inside the dense transition window

Because exactly the same prompt IDs are reused at each checkpoint, adjacent-checkpoint changes can be examined as paired flips. Aggregating the four tasks descriptively gives the following counts:

Transition	Wrong → correct	Correct → wrong	Net
600 → 700	62	67	-5
700 → 850	34	9	+25
850 → 900	26	7	+19
900 → 1000	28	12	+16

The largest directional change begins after step 700. At 700 → 850, fruit/month has 7 wrong-to-correct versus 0 correct-to-wrong cases (raw exact $p = 0.0156$) and furniture/profession has 6 versus 0 ($p = 0.0313$). At 850 → 900, four-class has 12 versus 3 ($p = 0.0352$). At 900 → 1000, fruit/month has 11 versus 0 ($p = 0.0010$). However, **none of the 64 individual task-by-adjacent-transition tests survives the notebook’s global Holm correction**; the smallest corrected value is 0.0625. These paired tests therefore support the direction and localization descriptively, but should not be presented as individually corrected-significant transitions.

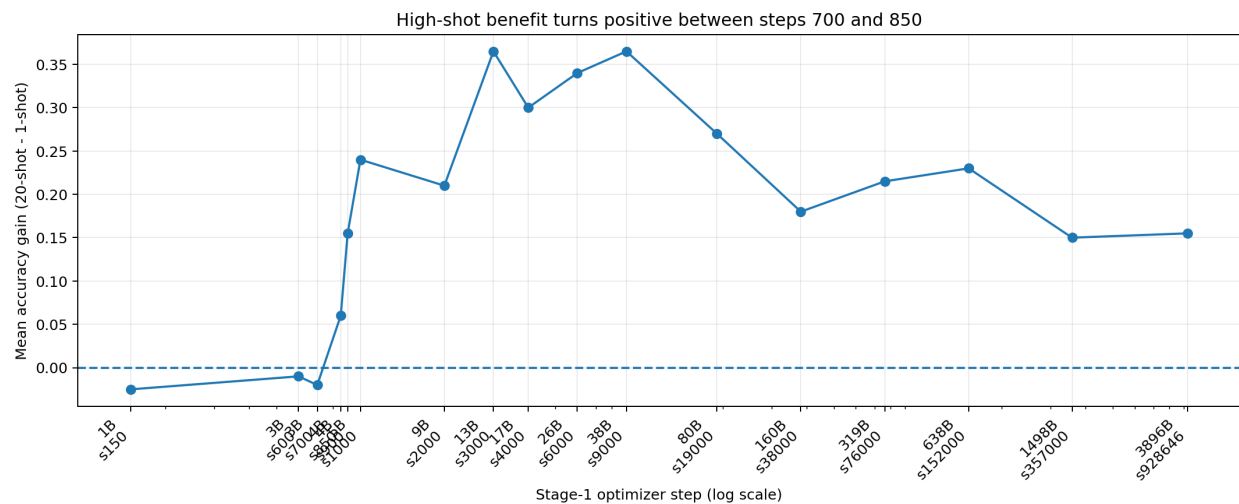
6. Development continues after onset, but is task-dependent



The shared early transition does not imply a shared peak. Fruit/month first has a Wilson interval fully above chance at step 850 and later reaches 0.94 at 17B / step 4000. Nine-class also first clears chance at step 850, but peaks much later at 0.60 at 38B / step 9000. Furniture/profession and four-class first clear chance at step 900; they peak at 0.84 at 5B / step 1000 and 0.86 at 13B / step 3000 respectively. This suggests a common early acquisition phase followed by task-specific refinement.

Task	First CI > chance	Peak checkpoint	Peak acc.	Max 20–1 gain
Fruit/month	4B / s850	17B / s4000	0.94	0.38 @ 17B / s4000
Furniture/prof.	4B / s900	5B / s1000	0.84	0.26 @ 5B / s1000
Four-class	4B / s900	13B / s3000	0.86	0.58 @ 38B / s9000
Nine-class	4B / s850	38B / s9000	0.60	0.48 @ 38B / s9000

7. High-shot benefit turns positive between steps 700 and 850



Mean 20-shot minus 1-shot accuracy gain is -0.020 at step 700, +0.060 at step 850, +0.155 at step 900, and +0.240 at step 1000. This provides a second behavioral localization that does not depend on task-specific chance levels: the demonstrations begin to produce a net high-context advantage between steps 700 and 850, and that advantage strengthens sharply by step 900.

8. The trajectory remains non-monotonic

Mean 20-shot accuracy across the four tasks is 0.320 at 1B / step 150, 0.360 at step 600, 0.335 at step 700, 0.460 at step 850, 0.555 at step 900, and 0.635 at step 1000. It later reaches 0.745 at 13B and a sampled maximum of 0.750 at 38B, then declines to 0.535 at the final 3896B checkpoint. The late decline occurs while mean format accuracy remains 1.00. A useful working description is therefore: **format acquisition** → **sharp early ICL transition near 4B** → **task-specific strengthening** → **partial later decline or reorganization**.

Tokens	Step	Mean acc.	Mean 20-1 gain	Format acc.
1B	150	0.320	-0.025	0.845
3B	600	0.360	-0.010	1.000
3B	700	0.335	-0.020	1.000
4B	850	0.460	+0.060	1.000
4B	900	0.555	+0.155	1.000
5B	1,000	0.635	+0.240	1.000
9B	2,000	0.600	+0.210	0.980
13B	3,000	0.745	+0.365	1.000
17B	4,000	0.675	+0.300	1.000
26B	6,000	0.705	+0.340	1.000
38B	9,000	0.750	+0.365	1.000
80B	19,000	0.665	+0.270	1.000
160B	38,000	0.580	+0.180	1.000
319B	76,000	0.590	+0.215	1.000
638B	152,000	0.615	+0.230	1.000
1498B	357,000	0.505	+0.150	1.000
3896B	928,646	0.535	+0.155	1.000

9. Completed RI sweep: design and primary measurements

The initial behavioral analysis fixed the broad developmental target *before* looking at RI: the strongest sampled boundary lay between 3B / step 600 and 5B / step 1000. The RI sweep sampled that window densely at step 700 / 3B, step 850 / 4B, and step 900 / 4B. After the RI analysis identified a source-routing maximum near step 850, we evaluated those same intermediate steps behaviorally; this follow-up localizes reliable cross-task ICL to step 900 while preserving the originally independent broad 600–1000 window.

The sweep uses 700 deterministically balanced AGENDA triplets (100 for each of the seven semantic relations) after excluding 60 verified source=target self-relations. All 1,024 OLMo attention heads are evaluated at every selected checkpoint. The primary developmental gate matches Ren et al.'s early-training rule: the source token must be the attention argmax, without requiring the strict dominance ratio. The original $\tau = 2.2$ criterion is stored in parallel as a sensitivity analysis.

For interpretation we keep three quantities separate:

$$\text{conditional RI} = \text{ri_argmax},$$

which measures tail-token promotion *given* a qualifying source-routing event;

$$\text{QK source-routing frequency} = \text{qk_argmax_frequency},$$

which measures how often the source is the attention argmax; and

$$\text{RI per opportunity} = \frac{\text{scored source-routing events}}{\text{all opportunities}} \times \text{conditional RI}.$$

The last identity holds numerically row-by-row in the saved results. It is important because an increase in RI-per-opportunity can arise from more frequent source routing even when conditional OV selectivity is unchanged.

10. RI result: source-directed routing peaks as behavioral ICL begins to rise

The clearest early RI-side event remains a transient increase in QK routing to the semantic source. Mean relaxed source-routing frequency rises from 0.0667 at step 600 to 0.0709 at step 700 and reaches a maximum of 0.0801 at step 850. It remains high at 0.0784 at step 900, then declines to 0.0719 at step 1000. The strict $\tau = 2.2$ routing frequency shows the same qualitative early increase, from 0.0145 at step 600 to 0.0216 at step 850.

The new behavioral interpolation makes the temporal relationship much more precise: mean 20-shot accuracy is still 0.335 at step 700, begins rising to 0.460 at the routing maximum (step 850), reaches 0.555 with all four task CIs above chance at step 900, and continues to 0.635 at step 1000 even as the routing frequency starts to fall.

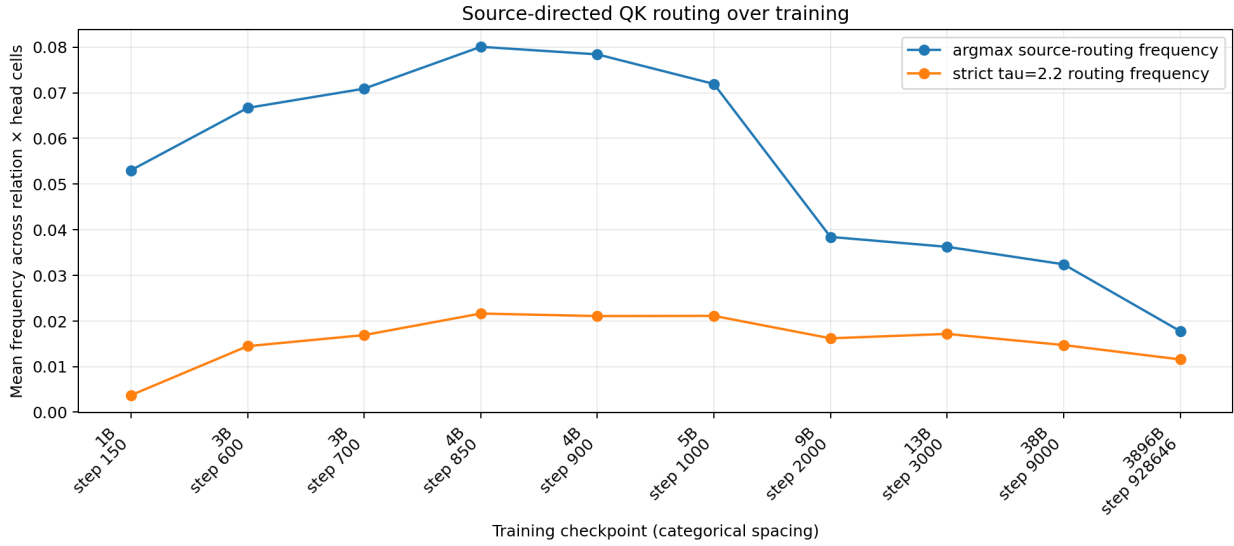


Figure 1: Population-average source-directed QK routing. The relaxed and strict gates strengthen sharply in the dense early window and peak near step 850.

Tokens	Step	ICL 20-shot	QK relaxed	QK strict	Cond. RI (bal.)	RI/opportunity
1B	150	0.320	0.0530	0.0037	0.0789	0.00415
3B	600	0.360	0.0667	0.0145	0.0831	0.00508
3B	700	0.335	0.0709	0.0169	0.0818	0.00533
4B	850	0.460	0.0801	0.0216	0.0805	0.00595
4B	900	0.555	0.0784	0.0211	0.0809	0.00583
5B	1,000	0.635	0.0719	0.0211	0.0812	0.00541
9B	2,000	0.600	0.0384	0.0162	0.0816	0.00302
13B	3,000	0.745	0.0362	0.0172	0.0831	0.00288
38B	9,000	0.750	0.0324	0.0147	0.0849	0.00267
3896B	928,646	0.535	0.0178	0.0116	0.0954	0.00167

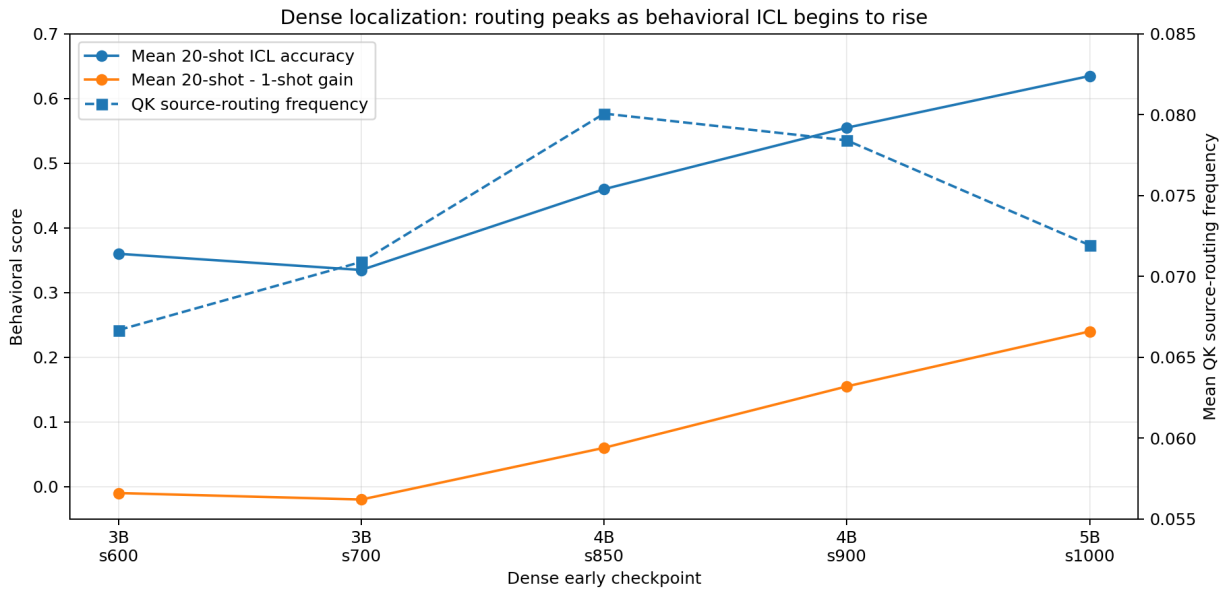


Figure 2: Dense 600–1000 follow-up. QK source routing reaches its maximum at step 850, when mean behavioral performance and high-shot benefit begin to rise; reliable cross-task behavior is established by step 900. The behavioral interpolation at 700/850/900 was a targeted follow-up and should be interpreted descriptively rather than as an independently preregistered timing test.

The relaxed source-routing increase from step 600 to step 850 is about 20% relative, while the strict-gate increase is about 49% relative. This places a substantial internal routing change immediately before or at the beginning of the localized behavioral transition. It is suggestive temporal alignment, but it is not evidence that the routing change causes ICL.

The early change is also relation-dependent. From step 600 to step 900, the largest relaxed-QK increases occur for

Compare (+0.0281), Used-for (+0.0220), Evaluate-for (+0.0150), and Part-of (+0.0123); Conjunction increases more modestly, Feature-of is approximately flat, and Hyponym-of decreases slightly. Thus the population effect should be described as a *relation-dependent routing transition*, not a universal simultaneous phase transition across all seven relations.

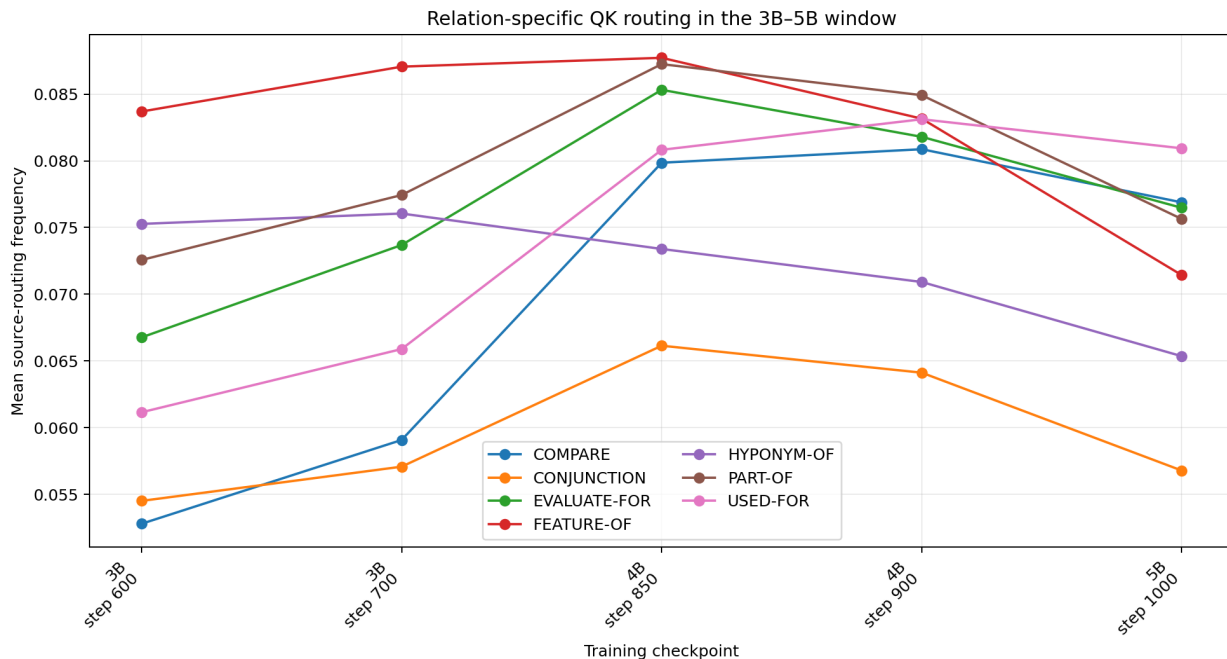


Figure 3: Relation-specific source-routing frequencies in the dense early window.

Interpretation: The strongest current temporal statement is: source-directed QK routing peaks at step 850, the first clear behavioral rise appears at the same checkpoint, and all four tasks meet the Wilson-above-chance criterion by step 900. Conditional RI does not rise with this event. Because the dense behavioral checkpoints were added after the RI peak was known, this exact alignment is a targeted follow-up observation; the broader 600–1000 alignment remains the independently established comparison.

11. Conditional RI itself does not show the same early jump

The metric most directly comparable to the paper's Relation Index is the conditional RI score after the source-routing gate is satisfied. Unlike QK frequency, its population mean is nearly flat throughout the dense 3B–5B window:

$$0.0831 (s600) \rightarrow 0.0818 (s700) \rightarrow 0.0805 (s850) \rightarrow 0.0809 (s900) \rightarrow 0.0812 (s1000).$$

It rises more visibly only later; the relation-balanced mean reaches about 0.0849 at 38B and 0.0954 at the final 3896B checkpoint.

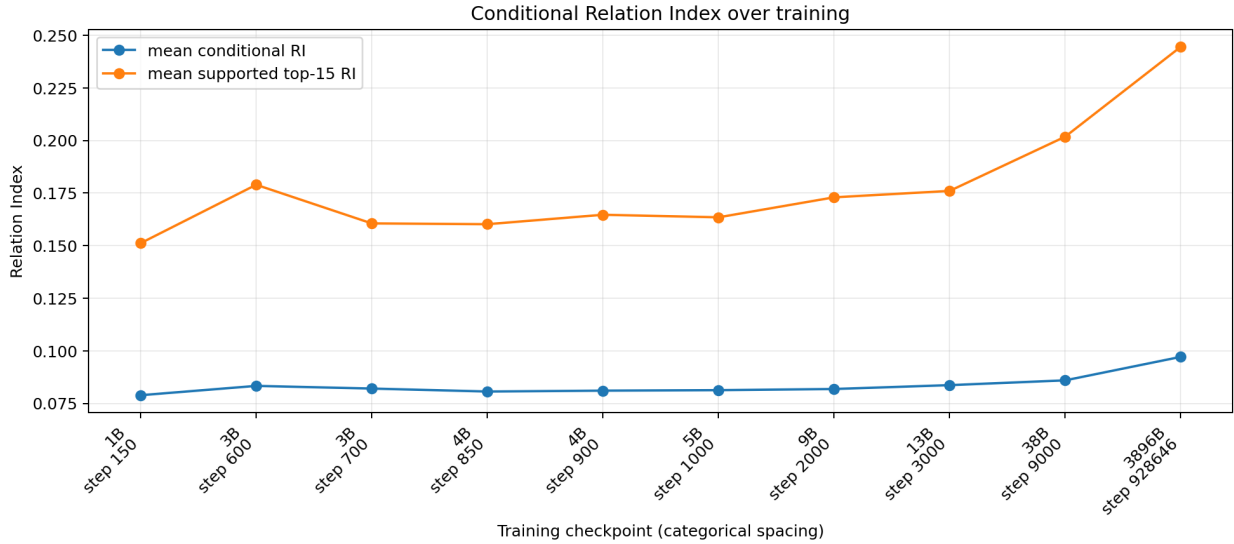


Figure 4: Population-average conditional RI and supported top-15 RI. Conditional RI does not exhibit a sharp increase across the localized step-700–900 behavioral transition.

This distinction changes the appropriate comparison with Ren et al. Their developmental claim is that the RI of *some heads* rises around the same stage as ICL. Our results do contain individual heads with increasing RI, but across the full population the conditional RI average does not show a common step-700–900 jump. Selecting only post-hoc increasing heads would therefore be vulnerable to selection effects. A closer paper-style follow-up should use a fixed, predeclared head-selection rule and trace those coordinates across checkpoints.

The combined RI-per-opportunity quantity peaks at 4B / step 850 because the source-routing term increases there. Since conditional RI is approximately stable, the early combined peak is primarily a routing-frequency effect.

12. Paired RI bootstrap confirms the routing/conditional-RI separation

We quantified uncertainty using a **paired cluster bootstrap** over the saved RI sweep chunks. The same frozen 20-text-group chunks were resampled with the same multiplicities at both endpoints of each comparison, preserving the pairing of AGENDA material across checkpoints. We used 5,000 bootstrap replicates and reconstructed each statistic from the additive chunk-level sufficient statistics. Heads are fixed model components and were not treated as independent bootstrap units. The overall statistic is relation-balanced: each of the seven semantic relations contributes equal weight.

Comparison	Δ QK	95% CI	Δ cond. RI	95% CI
600 $\rightarrow$ 850	+0.01339	[0.01010, 0.01633]	-0.00254	[-0.00347, -0.00098]
600 $\rightarrow$ 900	+0.01173	[0.00801, 0.01517]	-0.00217	[-0.00320, -0.00050]
600 $\rightarrow$ 1000	+0.00525	[0.00077, 0.00942]	-0.00189	[-0.00378, 0.00033]

The routing increase is therefore statistically robust under this resampling scheme. For step 600 $\rightarrow$ 850 and 600 $\rightarrow$ 900, every one of the 5,000 relation-balanced bootstrap replicates gives a positive QK change. For step 600 $\rightarrow$ 1000, 98.54% of replicates are positive and the percentile interval still lies above zero.

In the primary paired cluster bootstrap, conditional RI shows a small negative shift from step 600 to steps 850 and 900, while the step-1000 interval includes zero. This already rules out a positive population-wide RI jump in the same interval as the QK increase. However, conditional RI is defined only when a head has scored source-routing events, so the contributing head population can change as routing support changes. The fixed-support sensitivity analysis below isolates that issue and shows that much of the small negative shift is attributable to support composition rather than a robust within-head decrease.

The relation-specific intervals support the same interpretation. For both 600 $\rightarrow$ 850 and 600 $\rightarrow$ 900, Compare, Conjunction, Evaluate-for, Part-of, and Used-for have QK intervals entirely above zero, whereas Feature-of and Hyponym-of do not. No relation shows a 95% confidence interval supporting a positive conditional-RI increase. Instead, Evaluate-for and/or Part-of show reliable decreases in several comparisons. By step 1000, the broad routing effect has partially receded: Compare and Used-for remain clearly positive, Feature-of is negative, and several other relations overlap zero.

Interpretation: The paired bootstrap strengthens the central mechanistic conclusion. The step-850/900 event is a robust, relation-dependent increase in source-directed routing, not a simultaneous population-wide increase in conditional Relation Index. This is stronger evidence for an early QK-routing reorganization than for an early emergence of the mature RI/SIH signature.

Fixed-support sensitivity: conditional RI is stable within the same supported heads

To separate changes in conditional RI from changes in which heads have enough source-routing support, we performed a fixed-support sensitivity analysis. For each relation and endpoint pair, we defined the retained head set from the full non-bootstrap data only:

$$\mathcal{H}_{\text{fixed}} = \left\{ h : n_{\text{argmax_scored},h}^{(a)} \geq 10 \text{ and } n_{\text{argmax_scored},h}^{(b)} \geq 10 \right\}.$$

This set was frozen before uncertainty estimation and was selected only by support, never by the direction or magnitude of RI. We then recomputed conditional RI using exactly those same relation–head coordinates at both checkpoints.

A standard multinomial chunk bootstrap can occasionally omit every chunk containing the scored events of a low-support retained head, creating a zero denominator even though the full-data support criterion is satisfied. This occurred in roughly 80% of replicates when asking whether *at least one* of more than 5,000 retained relation–head cells became undefined. This does not mean that most heads were missing; it is a family-wise event across thousands of cells. To make the fixed-support requirement literal in every replicate, we therefore used a **paired Bayesian cluster bootstrap**: every 20-text-group chunk receives a strictly positive Dirichlet weight, and the same weights are used at both endpoints. Under this estimator, the zero-denominator rate is exactly 0 for all three comparisons.

Comparison	Retained relation–head cells	Δ fixed RI	95% Bayesian interval	$P(\Delta > 0)$
600 → 850	5,406	-0.000326	[-0.001176, 0.000452]	0.193
600 → 900	5,344	-0.000161	[-0.001094, 0.000693]	0.303
600 → 1000	5,205	+0.000951	[-0.000129, 0.001853]	0.952

All three 95% intervals include zero. In particular, the fixed-support estimates for 600 → 850 and 600 → 900 are extremely close to zero. The small negative changes seen in the primary population bootstrap therefore attenuate almost completely once the supported head population is held fixed. At 600 → 1000 the point estimate becomes mildly positive, but its interval still includes zero, so it is not treated as established RI growth.

The correct interpretation is consequently more precise than either “RI decreases” or “RI increases”: **within a fixed population of adequately supported heads, conditional OV tail selectivity is approximately stable across the early ICL transition.** What changes robustly is which heads route attention to the semantic source, and how often they do so.

Interpretation: The fixed-support sensitivity resolves the main remaining statistical ambiguity. The robust developmental event is the QK/source-routing reorganization. Conditional RI within the same adequately supported relation–head population remains approximately unchanged from step 600 through the step-850/900 ICL transition; the small negative shift in the unrestricted bootstrap is largely a support-composition effect.

13. Mature high-RI head identity is not stable at ICL onset

A second strong result is that the mature high-RI head population is not already present as a stable set during the localized early transition. For each relation, we selected the final checkpoint’s supported top-15 heads and compared earlier top-15 sets with that final set. Mean Jaccard overlap is 0.000 at 3B / step 600 and 4B / step 850, and only 0.0049 at 4B / step 900 and 5B / step 1000. All-head Spearman similarity to the final ranking is also low in the early window (about 0.13–0.15), increasing gradually to 0.33 by 38B.

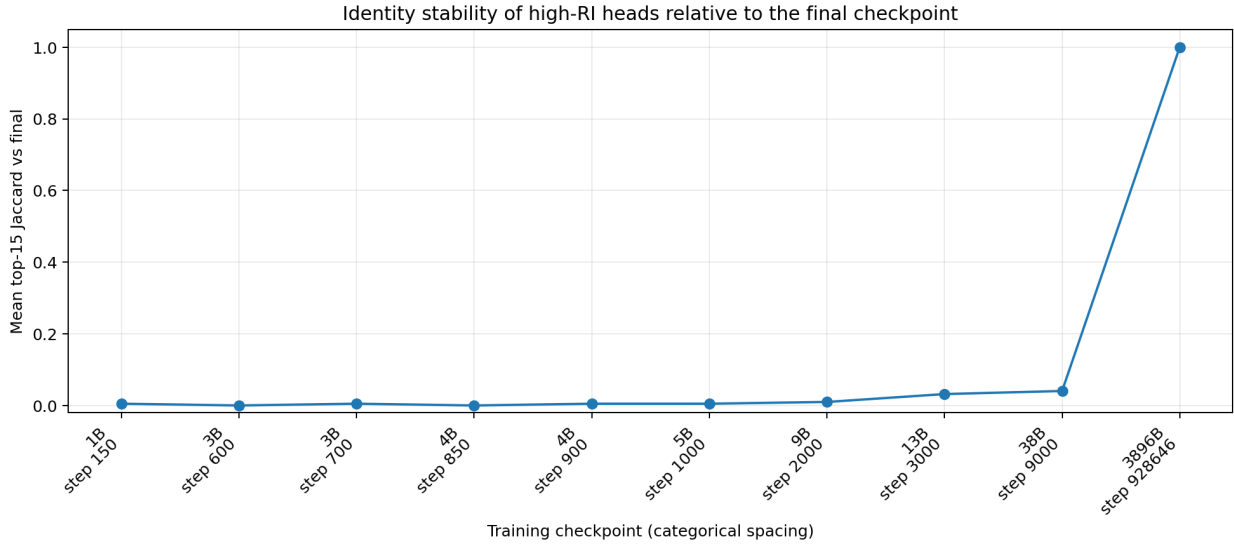


Figure 5: Mean relation-wise top-15 Jaccard overlap with the final high-RI head set. The mature head set is essentially absent as a stable top-ranked population during the early ICL transition.

There is some short-range stabilization inside the 4B window: mean adjacent-checkpoint top-15 Jaccard is about 0.31 from step 850 to 900 and 0.20 from step 900 to 1000. However, it falls to about 0.02 from step 1000 to 2000. This pattern is more consistent with transient early organization followed by substantial head-population turnover than with one fixed SIH set appearing once and remaining unchanged.

14. Final high-RI heads acquire strong semantic preference much later

Freezing the final top-RI head coordinates and tracing those same heads backward gives another perspective. Their mean conditional RI is only 0.0751 at 3B / step 600, 0.0698 at 4B / step 850, and 0.0751 at 5B / step 1000. It rises to 0.0934 at 13B, 0.1020 at 38B, and 0.2445 at the final checkpoint.

More importantly, their target-minus-random-visible-token gap is approximately zero or negative during the ICL onset window:

$$-0.0003 \text{ (3B/s600)}, \quad -0.0112 \text{ (4B/s850)}, \quad -0.0146 \text{ (4B/s900)}, \quad -0.0045 \text{ (5B/s1000)},$$

but becomes strongly positive at the final checkpoint (+0.1562).

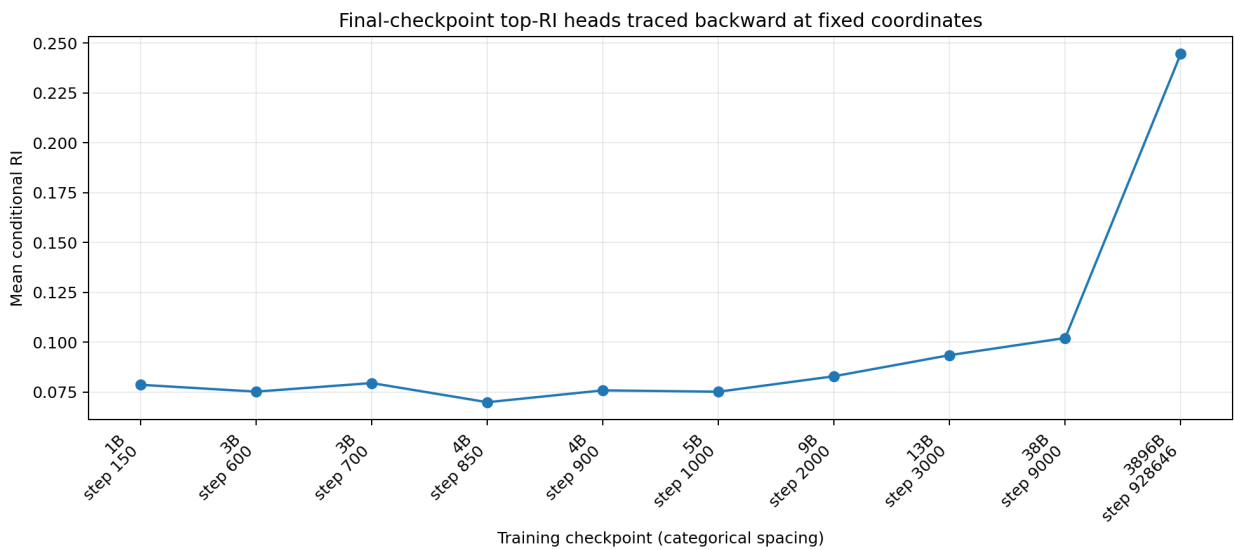


Figure 6: Conditional RI of the final checkpoint's high-RI heads traced backward at fixed coordinates.

This argues against the literal interpretation that the final model's mature high-RI heads already possessed their mature semantic target preference at the moment behavioral ICL first appeared.

The top-15 target-vs-random null comparison at each checkpoint is useful as a sanity check, but it is selection-biased

because those heads are chosen using target RI itself. The fixed-final-head backward trace above is therefore the more informative developmental control.

15. Relation to the previous single-hop mechanistic experiments

Seven of the 25 previously strong Stage-3 single-hop head coordinates occur in at least one relation's final AGENDA top-15 set:

$\{L17H1, L17H24, L18H19, L22H5, L27H6, L30H13, L30H18\}$.

This is an interesting exploratory cross-task overlap, but it does not overturn the earlier causal result. In the synthetic mother-of task, pooled RI was a weak causal ranking ($\rho = 0.135$), several high-RI candidates had negligible causal effects, and several major causal heads had low RI. The current AGENDA result instead says that some mature relationally active coordinates are shared across tasks.

The two findings are compatible:

a head can acquire a strong AGENDA relation signature

without

RI being a reliable causal-importance ranking for the mother-of task.

A formal overlap significance test should be layer-aware and treated as exploratory because both head sets were selected after substantial prior analysis.

16. Updated intermediate conclusions

The behavioral timing is now localized much more tightly. The original independent sweep bracketed the transition between step 600 (3B) and step 1000 (5B). The targeted intermediate evaluation shows that behavior remains weak at step 700, begins improving at step 850, and is reliable across all four tasks by step 900. Thus the best current description is a **step-700–900 transition centered near 4B training tokens**, with step 900 as the first sampled checkpoint at which all four 20-shot Wilson intervals lie above chance. This is strikingly close in token scale to Ren et al.'s approximately 4B-token pattern-discovery regime.

The mechanistic comparison remains a partial, not exact, replication. QK source routing reaches its maximum at step 850, exactly where the dense behavioral follow-up first shows a substantial rise, and it remains near that maximum at step 900 when cross-task ICL becomes robust. The paired cluster bootstrap confirms that the relation-balanced QK increase is positive for $600 \rightarrow 850$ ($\Delta = +0.01339$, 95% CI [0.01010, 0.01633]) and $600 \rightarrow 900$ ($\Delta = +0.01173$, 95% CI [0.00801, 0.01517]). The unrestricted bootstrap showed small negative conditional-RI shifts, but the fixed-support Bayesian sensitivity reduces those changes to -0.000326 (95% interval [-0.001176, 0.000452]) and -0.000161 (95% interval [-0.001094, 0.000693]), respectively. Thus, once the same adequately supported heads are compared, conditional RI is approximately stable. The early combined RI-per-opportunity maximum is therefore primarily a routing-frequency effect.

The most precise current statement is:

OLMo exhibits a transient, relation-dependent increase in source-directed attention routing that peaks at step 850, at the beginning of a behavioral ICL transition that is robust across tasks by step 900. Conditional Relation Index does not show a corresponding population-wide rise; under fixed-support Bayesian resampling it remains approximately stable within the same adequately supported heads. The mature high-RI head population is also not yet stable.

The exact 850/900 alignment should be interpreted with a design caveat. The broad step-600–1000 ICL boundary was measured before RI timing was interpreted, but the extra behavioral checkpoints 700/850/900 were added specifically after the RI sweep motivated closer inspection. Therefore the dense temporal match is valuable follow-up evidence, not a fully independent confirmation of a predeclared step-850 prediction.

The result remains compatible with the previous Stage-1/2/3 causal work. Those experiments showed that RI is a poor mature causal selector for the synthetic mother-of task; the present side experiment instead studies when routing and RI signatures develop. Developmental temporal association and mature causal ranking are distinct hypotheses.

Late OLMo dynamics remain distinct from Ren et al. Behavioral accuracy is non-monotonic and declines late despite preserved format compliance, while conditional RI and the final-head semantic preference continue to strengthen. This dissociation suggests that mature RI-style specialization is not sufficient to preserve peak performance on these arbitrary few-shot mappings.

17. Optional extensions and stopping point

The main developmental question is now statistically well supported: dense behavioral checkpoints localize ICL acquisition, the paired QK bootstrap is complete, and the fixed-support Bayesian sensitivity shows that the same adequately supported heads do not exhibit a concurrent conditional-RI jump. For a course/project-scale study, this is a reasonable stopping point for the developmental side experiment. The items below are optional extensions rather than required validation checks.

1. **Paper-fidelity sensitivity at key checkpoints.** If a closer replication is desired, create a separately named stream using Ren et al.'s spaCy frequent-function-word removal and evaluate the reverse semantic direction. Restrict this to steps 600, 850, 900, 1000, and 928646 to control compute.
2. **Reproduce the paper's head-grouping analysis more literally.** For each triplet, identify the head with the largest positive averaged RI, count head occurrences by relation, freeze the resulting high-occurrence set, and trace those coordinates backward. This directly tests the paper's "some heads increase" visualization without selecting trajectories merely because they rose post hoc.
3. **Balanced-subsample robustness only if additional assurance is needed.** At steps 600/850/900/1000, repeat the RI analysis with several deterministic alternative 100-per-relation manifests or all eligible rows. This would test dependence on one balanced draw without rerunning the entire trajectory.
4. **Layer-aware overlap test for the previous causal heads.** Preserve layer structure when drawing comparison sets because both mature RI and causal effects are layer-concentrated. Treat this as exploratory.
5. **Causal intervention as a new research question.** If the project is extended beyond temporal association, identify heads contributing to the early routing increase and ablate or patch them at steps 850/900 while measuring the newly acquired ICL behavior. This would test mechanistic involvement rather than correlation, but it is a distinct experiment rather than a necessary robustness check.

Interpretation: The developmental side experiment can reasonably be frozen here. The central result survives the main statistical checks: QK routing changes robustly around ICL acquisition, while fixed-support conditional RI remains approximately stable. Further work would improve replication fidelity or test causality, but is not required to support the current conclusion.

18. Current status

Component	Status
Frozen ICL stream + integrity audit	Complete
Frozen AGENDA / OLMo-safe RI stream	Complete
17-checkpoint behavioral sweep	Complete, including steps 700/850/900
Behavioral analysis	Dense transition localized with exact step identity
Behavioral onset estimate	Begins after step 700; reliable across all four tasks by step 900 (approximately 4B)
Developmental RI sweep	Complete: 10 checkpoints, 700 balanced triplets, all 1,024 heads
RI analysis notebook	Complete; paired chunk bootstrap and fixed-support Bayesian sensitivity complete
Early RI-side result	Robust QK increase; fixed-support conditional RI is approximately stable at 850/900
Mature-head stability	Final top-RI coordinates are not stable during early ICL onset
Comparison with Ren et al.	Behavioral timing is very close; mechanistic result is partial alignment at QK routing, not a broad conditional-RI replication
Comparison with prior causal RI work	Compatible: developmental timing and mature causal ranking are distinct questions
Remaining work	No required statistical check; paper-fidelity or causal extensions are optional

Bottom line: the dense behavioral follow-up sharpens the side experiment substantially. OLMo remains weak through step 700, begins improving at step 850, and shows reliable 20-shot behavior across all four tasks by step 900, placing the main ICL transition near 4B training tokens. Source-directed QK routing peaks at step 850 and remains elevated at step 900. The paired 5,000-replicate chunk bootstrap confirms that the relation-balanced QK increase is robust for 600 → 850 and 600 → 900. The final fixed-support paired Bayesian cluster bootstrap eliminates zero-denominator support changes

and finds conditional-RI changes compatible with zero at 600 → 850 (-0.000326 , 95% interval $[-0.001176, 0.000452]$) and 600 → 900 (-0.000161 , 95% interval $[-0.001094, 0.000693]$); the small positive estimate at 600 → 1000 also has an interval crossing zero. The current evidence therefore supports **a temporally aligned early routing reorganization**, but only **partial support for a broad early SIH/RI formation account**. The exact step-level alignment is targeted follow-up evidence rather than an independently preregistered timing test. This completes the main statistical robustness checks for the developmental side experiment. The result remains fully compatible with the earlier finding that RI is not a reliable causal head-ranking metric for the mature single-hop mother-of task.